

**Sentiment Classification of IMDb Reviews Using a Long Short-Term Memory
Network**

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Sentiment analysis is a common natural language processing task in which a system determines whether a text expresses positive or negative opinion. In this lab, the objective was to classify IMDb movie reviews into binary sentiment categories. This problem is sequence-sensitive because the sentiment of a sentence often depends on the order in which words appear rather than on isolated tokens alone.

A standard feedforward architecture is limited for this purpose because it does not naturally preserve contextual ordering. Long short-term memory networks are more appropriate because they process sequences while retaining relevant information through gated memory. This makes them effective for phrases such as `not good`, where the final meaning depends on context.

Scenario

The working scenario for this lab is the development of an automated review analysis system that can identify the sentiment of movie reviews as positive or negative. Because movie reviews are sequential text, the model must be able to understand contextual order rather than relying only on isolated words.

Task 1: Data Preparation

The IMDb movie review dataset used in this experiment contains 50,000 labeled reviews. Before model training, the reviews were cleaned, tokenized, padded, and split into training, validation, and testing subsets. The cleaning stage removed HTML tags, normalized whitespace, retained only relevant characters, and converted all tokens to lowercase.

Table 1*Dataset and Preprocessing Configuration*

Item	Configuration
Dataset Source	IMDB Dataset.csv containing 50,000 movie reviews
Vocabulary Limit	5,000 tokens
Maximum Sequence Length	100 tokens
Train / Validation / Test Split	40,000 / 5,000 / 5,000
Padding and Truncation	Post padding and post truncation
Label Mapping	Negative = 0, Positive = 1

Code Example 1*Cleaning and Sequence Padding Procedure*

```
def clean_review(text):
    text = html.unescape(str(text))
    text = re.sub(r"<[^>]+>", " ", text)
    text = text.replace("\n", " ")
    text = re.sub(r"^[a-zA-Z0-9' ]+", " ", text)
    text = re.sub(r"\s+", " ", text).strip().lower()
    return text

tokenizer = Tokenizer(num_words=5000, oov_token="<OOV>")
tokenizer.fit_on_texts(train_df["review_clean"])

X_train = pad_sequences(train_sequences, maxlen=100,
                        padding="post", truncating="post")
```

Padding is necessary because text sequences vary naturally in length, whereas mini-batch training requires all inputs in a batch to share the same shape. Selecting a

fixed sequence length also improves efficiency and keeps the input representation consistent across the experiment.

Figure 1

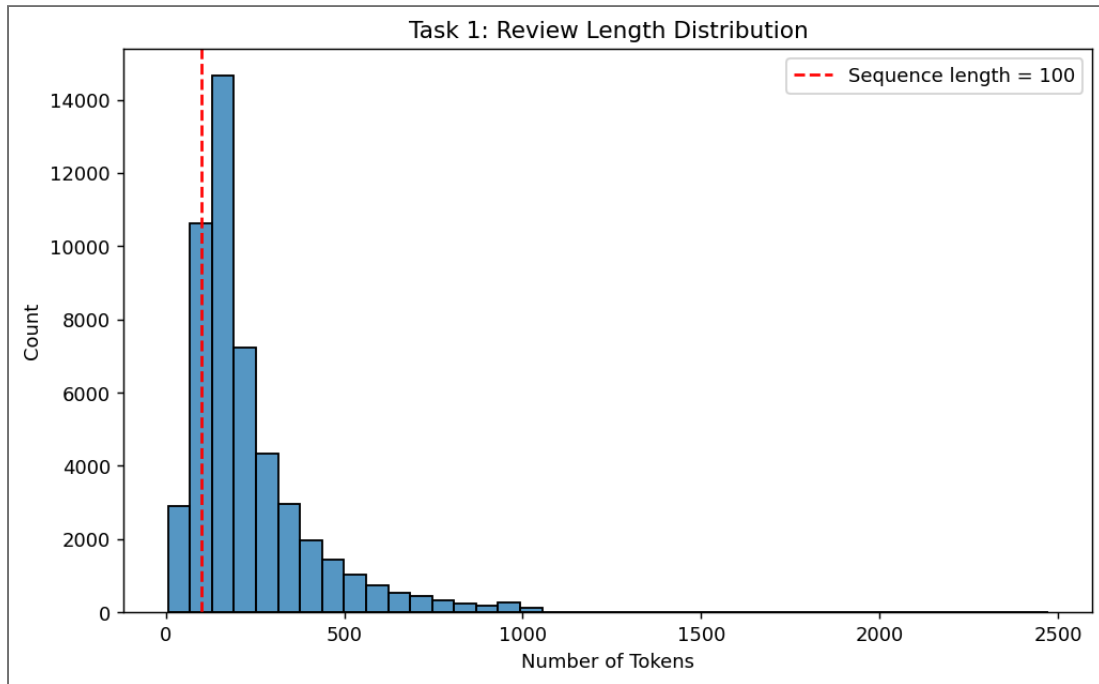
Sample normalized reviews after the cleaning stage

Task 1: Sample IMDb Reviews
Sentiment: Positive i really liked this summerslam due to the look of the arena the curtains and just the look overall was interesting to me for some reason anyways this could have been one of the best summerslam's ever if the wwf didn't have lex luger in the main event against yokozuna now for it's time it was ok to have a huge fat man vs a strong man but i'm glad times have changed it was a terrible main event just like every ...
Sentiment: Positive not many television shows appeal to quite as many different kinds of fans like farscape does i know youngsters and 30 40 years old fans both male and female in as many different countries as you can think of that just adore this t v miniseries it has elements that can be found in almost every other show on t v character driven drama that could be from an australian soap opera yet in the same episode it has ...
Sentiment: Negative the film quickly gets to a major chase scene with ever increasing destruction the first really bad thing is the guy hijacking steven seagal would have been beaten to pulp by seagal's driving but that probably would have ended the whole premise for the movie it seems like they decided to make all kinds of changes in the movie plot so just plan to enjoy the action and do not expect a coherent plot turn any sense of ...
Sentiment: Positive jane austen would definitely approve of this one gwyneth paltrow does an awesome job capturing the attitude of emma she is funny without being excessively silly yet elegant she puts on a very convincing british accent not being british myself maybe i'm not the best judge but she fooled me she was also excellent in sliding doors i sometimes forget she's american also brilliant are jeremy northam and sophie ...
Sentiment: Negative expectations were somewhat high for me when i went to see this movie after all i thought steve carell could do no wrong coming off of great movies like anchorman the 40 year old virgin and little miss sunshine boy was i wrong i'll start with what is right with this movie at certain points steve carell is allowed to be steve carell there are a handful of moments in the film that made me laugh and it's due almost ...

Note. These examples were used to visually verify that the review text had been normalized correctly before tokenization.

Figure 2

Distribution of review lengths in tokens



Note. The selected maximum sequence length of 100 tokens provides a balance between contextual coverage and computational cost.

Task 2: Model Construction

The baseline model used in this lab consists of an embedding layer, a single LSTM layer, and a sigmoid output neuron. The embedding layer converts token identifiers into dense word vectors, the LSTM captures sequential dependencies, and the sigmoid unit outputs the probability of positive sentiment.

Code Example 2

Baseline and Configurable LSTM Model Definitions

```
model = tf.keras.Sequential([
    tf.keras.layers.Embedding(input_dim=5000,
                              output_dim=64,
                              input_length=100),
    tf.keras.layers.LSTM(64),
    tf.keras.layers.Dense(1, activation='sigmoid')
])

def build_lstm_model(max_len, lstm_units=64,
                    dropout_rate=0.0, learning_rate=1e-3):
    model = tf.keras.Sequential([
        tf.keras.layers.Embedding(input_dim=MAX_WORDS,
                                  output_dim=EMBEDDING_DIM,
                                  input_length=max_len),
        tf.keras.layers.LSTM(lstm_units, dropout=dropout_rate),
        tf.keras.layers.Dense(1, activation="sigmoid"),
    ])

    model.compile(optimizer=tf.keras.optimizers.Adam(learning_rate),
                  loss="binary_crossentropy",
                  metrics=["accuracy"])

    return model
```

LSTM was selected instead of SimpleRNN because its gated structure is better at retaining important contextual information over longer sequences. This property is especially useful in sentiment analysis, where negation, contrast, and emphasis often change the meaning of a sentence.

Task 3: Training and Evaluation

Baseline Model Performance

The original model was trained for five epochs with batch size 32. The results indicate effective learning on the training set, but the validation metrics reveal a modest generalization gap.

- Training accuracy: 0.8817
- Validation accuracy: 0.8142
- Training loss: 0.2873
- Validation loss: 0.4262
- Test accuracy: 0.8164
- Test loss: 0.4230

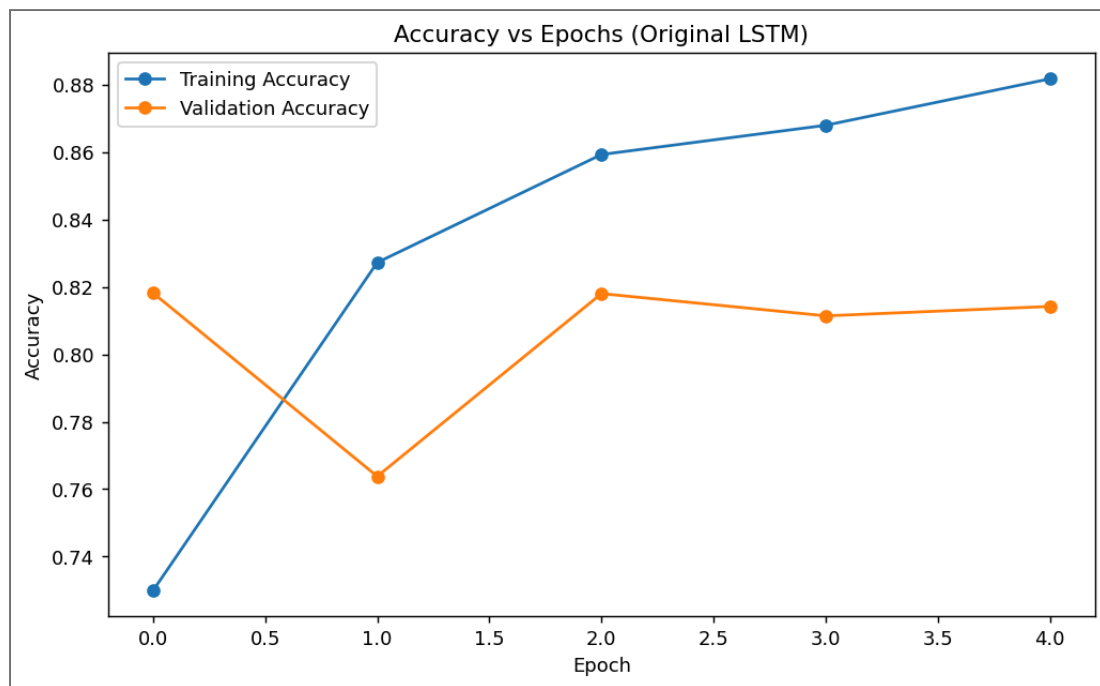
Code Example 3

Baseline Training Procedure

```
history_original = original_model.fit(  
    X_train_original,  
    y_train,  
    validation_data=(X_val_original, y_val),  
    epochs=5,  
    batch_size=32,  
    verbose=1,  
)  
  
original_test_loss, original_test_acc = original_model.evaluate(  
    X_test_original, y_test, verbose=1  
)
```

Figure 3

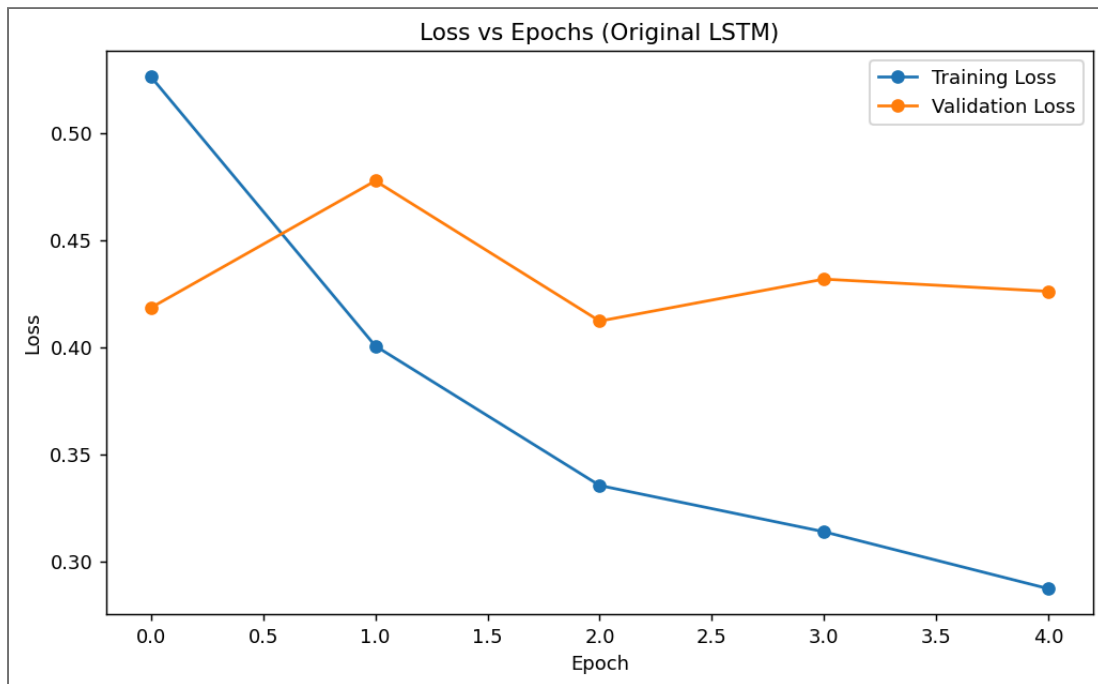
Training and validation accuracy for the baseline model



Note. The validation curve remains below the training curve, which suggests limited generalization relative to training performance.

Figure 4

Training and validation loss for the baseline model



Note. The consistently higher validation loss supports the interpretation of mild overfitting.

Task 4: Parameter Modification

Modified Model Performance

To improve performance, the number of LSTM units was increased from 64 to 128 and a dropout rate of 0.3 was introduced. The revised model aimed to combine stronger representational capacity with regularization.

Code Example 4

Updated LSTM Configuration

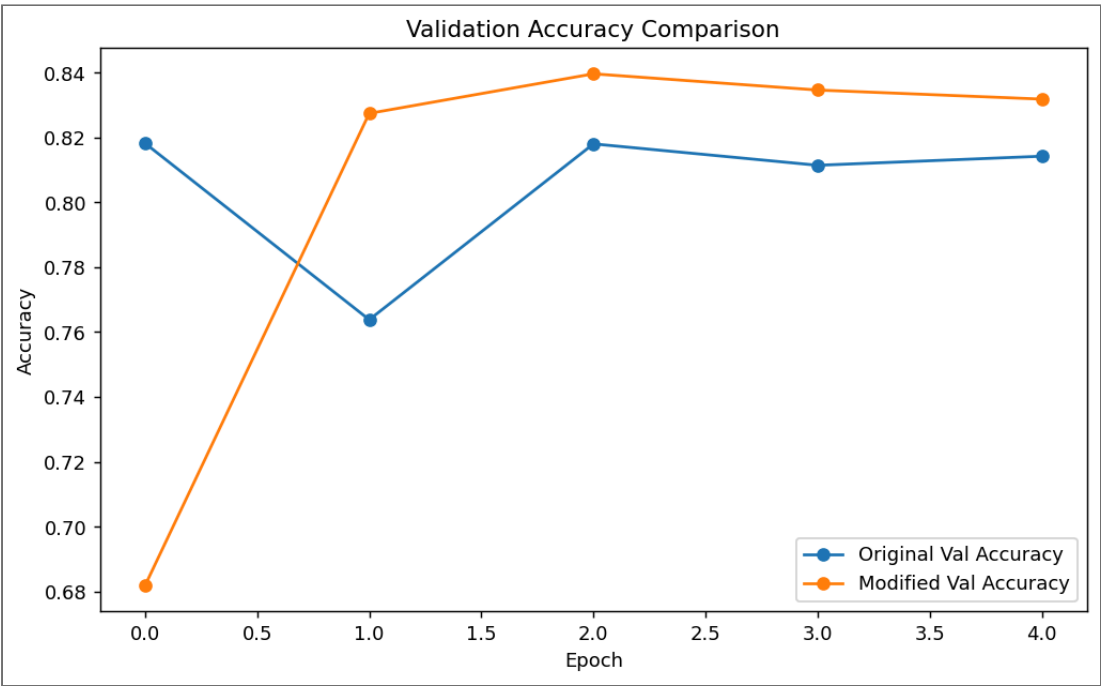
```
modified_model = build_lstm_model(  
    max_len=100,  
    lstm_units=128,  
    dropout_rate=0.3,  
)
```

Table 2
Comparison of Baseline and Modified Models

Model	Train Acc.	Val. Acc.	Test Acc.	Train Loss	Val. Loss	Test Loss
Original	0.8817	0.8142	0.8164	0.2873	0.4262	0.4230
Modified	0.8826	0.8318	0.8318	0.2818	0.3979	0.4001

The modified model improved both validation and test accuracy to 0.8318. This suggests that the revised configuration achieved better out-of-sample generalization rather than simply memorizing the training set more strongly.

Figure 5
Validation accuracy comparison for the two LSTM configurations



Note. The modified architecture produced stronger validation behavior across the training period.

Task 5: Final Prediction Analysis

Custom Prediction and Test Review

The modified LSTM was selected as the final model and evaluated on custom review samples as well as the held-out test set.

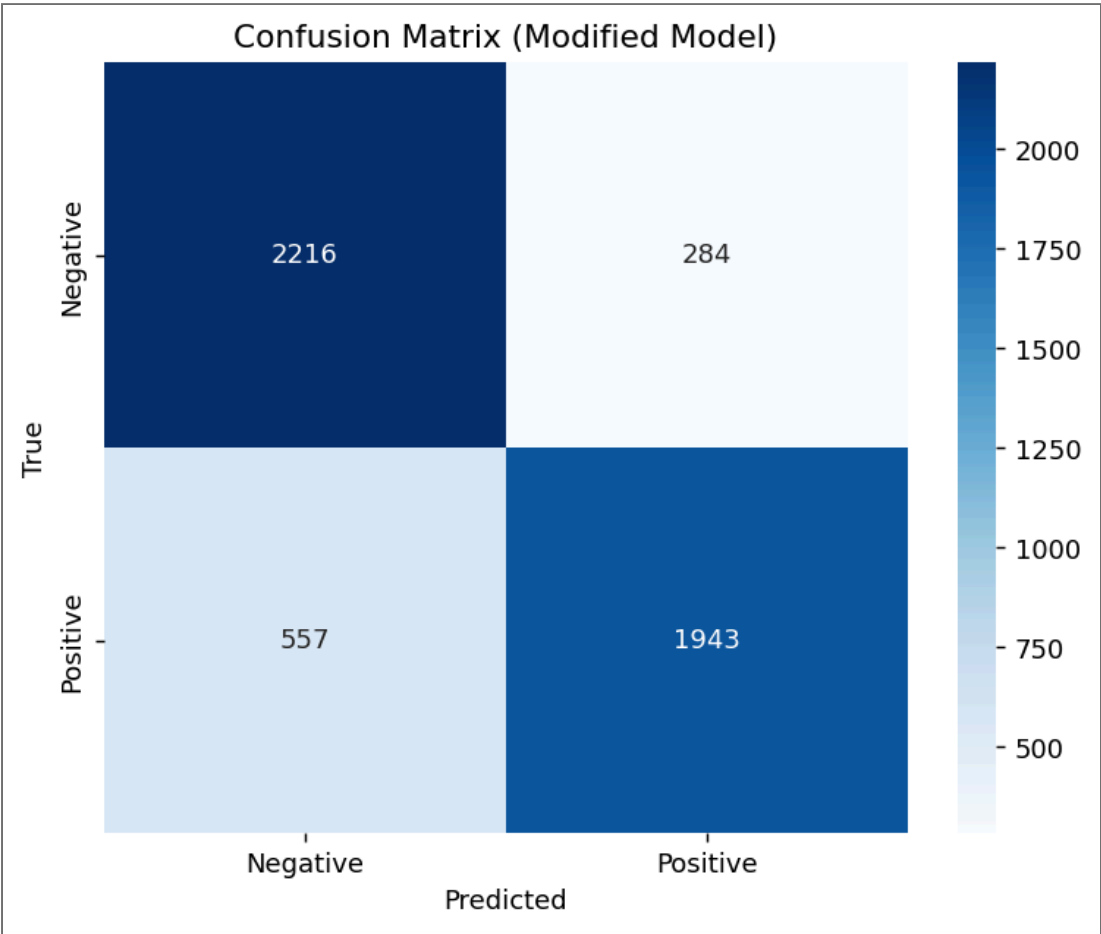
Table 3

Predictions for Custom Review Samples

Review	Predicted Probability	Predicted Sentiment
This movie was absolutely amazing	0.9819	Positive
This movie was not good at all	0.2619	Negative
The acting was great but the story was too slow and boring	0.0548	Negative
I really loved the characters and the ending was satisfying	0.9863	Positive
I expected something better, and honestly it was a waste of time	0.0165	Negative

Figure 6

Confusion matrix for the final model



Note. The final model showed balanced but slightly stronger recognition of negative reviews.

Code Example 5

Classification Report for the Final Model

	precision	recall	f1-score	support
Negative	0.7991	0.8864	0.8405	2500
Positive	0.8725	0.7772	0.8221	2500
accuracy			0.8318	5000
macro avg	0.8358	0.8318	0.8313	5000
weighted avg	0.8358	0.8318	0.8313	5000

Discussion

The findings of this lab show that LSTM is better suited than SimpleRNN for sentiment analysis because it can preserve useful context across longer sequences. The memory cell allows the network to connect earlier and later words, which is necessary for phrases in which negation changes polarity. Padding was also a necessary preprocessing step because variable-length reviews cannot be trained in batches without converting them to a common shape. Increasing the number of hidden units gave the model greater representational capacity, and the addition of dropout improved its ability to generalize.

Conclusion

This lab demonstrated the use of a long short-term memory network for binary sentiment analysis on IMDB movie reviews. The workflow included review cleaning, tokenization, padding, baseline model training, architectural modification, and final performance evaluation. The revised model outperformed the baseline and achieved 83.18% test accuracy. Overall, the experiment confirms that sequence-aware neural models are effective for text classification tasks in which context strongly influences meaning.

References

- Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. MIT Press.
- Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735-1780.
- Maas, A. L., Daly, R. E., Pham, P. T., Huang, D., Ng, A. Y., & Potts, C. (2011). Learning word vectors for sentiment analysis. In *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies* (pp. 142-150).